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LAB 5: IP Address Analysis and Network Report

Step 1: Select 10 Websites

1. Choose **10 popular websites** (e.g., apple.com, twitter.com, etc.) to analyze.
2. **Bonus Exercise:** Include a variety of domains, such as .com, .org, .edu, and country-specific domains (e.g., .co.uk, .de). Reflect on the differences in the IP address ranges assigned to these domains.

The 10 popular websites selected are:

Websites	IP Address	Class (A, B, or C)
Apple.com	17.253.144.10	A
twitter.com	104.244.42.193	A
google.com	142.250.200.142	B
youtube.com	142.250.200.78	B
Facebook.com	102.132.101.35	A
Wikipedia.org	185.15.58.224	B
Harvard.edu	192.0.66.20	C
researchgate.net	104.18.41.9	A
academia.edu	108.157.78.122	A

The above popular websites were ping and blow is the screenshot.

```
Engr.ABD@kali: ~
File Actions Edit View Help
rtt min/avg/max/mdev = 133.077/141.225/154.755/5.885 ms

(Engr.ABD@kali)-[~]
$ ping academia.edu
PING academia.edu (108.157.78.122) 56(84) bytes of data.
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=1 ttl=243 time=45.7 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=2 ttl=243 time=37.9 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=3 ttl=243 time=37.2 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=4 ttl=243 time=36.4 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=5 ttl=243 time=43.3 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=6 ttl=243 time=33.0 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=7 ttl=243 time=53.8 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=8 ttl=243 time=48.9 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=9 ttl=243 time=45.8 ms
64 bytes from server-108-157-78-122.los50.r.cloudfront.net (108.157.78.122): icmp_seq=10 ttl=243 time=49.3 ms
^C
  -- academia.edu ping statistics --
10 packets transmitted, 10 received, 0% packet loss, time 13097ms
rtt min/avg/max/mdev = 32.956/43.122/53.832/6.415 ms

(Engr.ABD@kali)-[~]
$ ping google.com
PING google.com (142.250.200.142) 56(84) bytes of data.
64 bytes from mad41s14-in-f14.1e100.net (142.250.200.142): icmp_seq=1 ttl=110 time=129 ms
64 bytes from mad41s14-in-f14.1e100.net (142.250.200.142): icmp_seq=2 ttl=110 time=128 ms
64 bytes from mad41s14-in-f14.1e100.net (142.250.200.142): icmp_seq=3 ttl=110 time=128 ms
64 bytes from mad41s14-in-f14.1e100.net (142.250.200.142): icmp_seq=4 ttl=110 time=122 ms
^C
  -- google.com ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 122.298/127.020/129.139/2.746 ms

(Engr.ABD@kali)-[~]
$
```

Step 4: Classify the IP Addresses

4. Determine the **class** (A, B, or C) of each IP address based on its first octet.

Here are the ranges for each class:

- **Class A:** 0–127/8 24
- **Class B:** 128–191/16 16
- **Class C:** 192–223/24 8

Bonus Exercise:

Research the reserved IP ranges (e.g., private IP ranges like 10.0.0.0/8) and see if any of your IP addresses fall into these special ranges.

The reserved IP addresses are the address used for internal network or private network within an organization to share or transfer file within an organization, if such IP address is exposed to internet, it could be a threat to the organization assets.

Here are the range of the reserved IP addresses:

Class A 10.0.0.0 – 10.255. 255. 255\8

Class B 172.16.0.0 – 172.31. 255. 255\8

Class C 192. 168.0.0 – 192. 168.255. 255\16

None of the IP addresses fall in to the range of popular websites above.

Discuss the importance of these reserved IPs in network configurations.

maintenance of routing tables, operational resilience during failures, and facilitating multicast communications. These addresses help in managing network resources effectively and prevent conflicts that could arise from overlapping addresses in dynamic environments.

It's important to verify specific details about reserved IPs with reliable sources to ensure accurate and up-to-date information.

Step 5: Calculate the Number of Devices Each IP Can Support

5. For each IP class, calculate how many devices the network can accommodate. This will give you an understanding of network size and scalability.

Device Capacity Formula:

- **Class A:** $2^{24} - 2$
- **Class B:** $2^{16} - 2$
- **Class C:** $2^8 - 2$

Bonus Exercise:

Calculate the actual number of usable hosts by performing subnetting on these networks. For instance, consider subdividing a Class B network into smaller subnets and calculate the device capacity for each subnet.

To calculate the usable hosts in a subnet, you first need to determine the subnet mask and the number of bits allocated for host addresses. For a Class B network, which has a default subnet mask of 255.255.0.0 (or /16), you can subnet further to accommodate specific requirements.

For example, if you choose to create 256 subnets, you will need an additional 8 bits (since $2^8 = 256$) for subnetting, leading to a subnet mask of 255.255.255.0 (or /24). In each /24 subnet, the formula for calculating usable hosts is:

$$\text{Usable Hosts} = 2^{(\text{Total Host Bits})} - 2$$

In a /24 subnet, there are 8 bits remaining for hosts:

$$\text{Usable Hosts} = 2^8 - 2 = 256 - 2 = 254$$

Thus, each /24 subnet can support 254 usable hosts.

It's essential to remember to subtract 2 from the total host count for the network and broadcast addresses, which cannot be assigned to hosts. Always double-check your calculations for accuracy.

- **Class A:** $2^{24} - 2 = 16,777,214$
- **Class B:** $2^{16} - 2 = 65,536$
- **Class C:** $2^8 - 2 = 254$

Consider subnet masks like /26 or /28 for Class C networks and explore how they impact the number of devices.

Subnet masks determine how many devices can be connected on a network by defining the size of the subnet. For Class C networks:

- A /26 subnet mask allows for 64 total addresses ($2^{(32-26)}$), which includes 62 usable addresses for devices (subtracting 1 for the network address and 1 for the broadcast address).
- A /28 subnet mask allows for 16 total addresses, resulting in 14 usable addresses for devices under the same subtraction rule.

Thus, using a /26 subnet offers more room for devices than a /28 subnet. **Class C 192.168.0.0/24**

Step 6: Determine the Subnet Mask

6. Identify the **default subnet mask** for each IP address based on its class.

Bonus Exercise:

Explore scenarios where custom subnetting is required (e.g., for network segmentation). How does altering the subnet mask affect the device capacity and overall network organization?

Class A = 255.0.0.0

Class B = 255.255.0.0

Class C = 255.255.255.0

Try experimenting with subnet masks like /25, /27 and explain the difference in the number of networks and hosts.

A /25 subnet mask allows for 128 IP addresses in total, with 126 usable for hosts (1 for the network address and 1 for the broadcast address). This provides 2 subnets, each supporting up to 126 hosts.

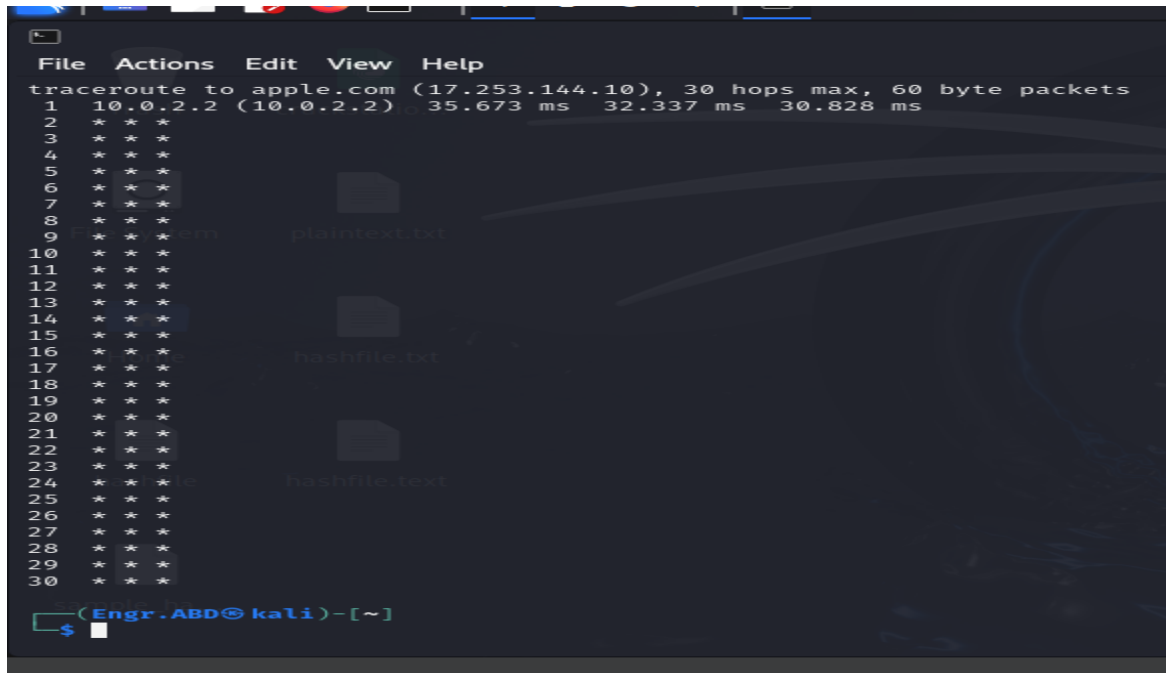
A /27 subnet mask provides 32 total IP addresses, with 30 usable for hosts (1 for the network and 1 for the broadcast). This can create 8 subnets, each supporting 30 hosts.

In summary:

- /25: 2 networks, 126 hosts each
- /27: 8 networks, 30 hosts each

Step 7: Advanced IP Tools (Bonus Exercises)

- **Traceroute (Network Path Analysis):** Use traceroute (Linux/Mac) or tracert (Windows) to discover the network path between your system and one of the websites.
- Analyze the hops between your system and the destination. Identify the location and IP address of each hop.

A screenshot of a terminal window on a Kali Linux system. The terminal shows the command 'traceroute to apple.com (17.253.144.10), 30 hops max, 60 byte packets'. The output displays 30 hops, with the first hop being 10.0.2.2 (10.0.2.2) with a round-trip time of 35.673 ms. The subsequent hops are marked with asterisks, indicating that the traceroute did not receive a response from those hops. The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The prompt at the bottom is '(Engr.ABD@kali)-[~]'.

GeoIP Location: Use online tools like IPinfo.io or GeoIP to check the geographical location of the IP addresses.

Identify if the IP addresses belong to data centers, ISPs, or any specific regions.

- ip:"17.253.144.10",
- hostname:"advertising.apple.com",
- city:"Cupertino",
- region:"California",
- country:"US",
- loc:"37.3230,-122.0322",
- org:"AS714 Apple Inc.",
- postal:"95014",
- timezone:"America/Los_Angeles",
- is_anycast:true,
- is_mobile:false,
- is_anonymous:false,
- is_satellite:false,
- is_hosting:false,
- asn:Object,
 - asn:"AS714",
 - name:"Apple Inc.",
 - domain:"apple.com",
 - route:"17.253.144.0/24",
 - type:"business",
- company:Object,
 - name:"Apple Inc.",
 - domain:"apple.com",
 - type:"business",
- privacy:Object,
 - vpn:false,

- proxy:false,
- tor:false,
- relay:false,
- hosting:false,
- service:"",
- abuse:Object,
 - address:"US, CA, Cupertino, MS 89-1DR, 1 Infinite Loop, 95014",
 - country:"US",
 - email:"abuse@apple.com",
 - name:"Apple Abuse",
 - network:"17.0.0.0/8",
 - phone:"+1-408-974-7777",

Bonus: Compare the latency (ping time) with the geographic distance.

There is a strong correlation between geographic distance and latency (ping time), as the farther data must travel, the longer it typically takes. However, this relationship is not absolute; other factors, such as the number of hops between routers, network congestion, and the type of connection (e.g., fiber, satellite) can also significantly impact latency. Thus, while distance is a key factor, it is not the only one affecting ping times.

Shorter distances result in lower ping times

For example, a gamer 10 miles from a server will have a better ping rate than someone 100 miles away.